Norsat
International Inc.

USER MANUAL

Single-speed Integrated Antenna Control System
ACS240KU-A

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Revision History

REVISION	DESCRIPTION
A	Initial Release

1. System Introduction

Single-speed Integrated Antenna Control System is a new generation of antenna control products that integrates antenna control unit and antenna drive as a whole. It adopts the current mainstream industrial communication technology and high-reliability electronic components. At the same time, it uses a high-resolution color touch screen as the human-computer interaction interface, and has functions of position guidance, satellite guidance, automatic tracking, position guide and scanning, which also can realize antenna remote control through Ethernet interface.

Main features:

1) Integrate antenna control unit, drive unit and beacon tracking unit.
2) New humanized system menu design and operation experience.
3) The high-grade switches, indicator lights and aviation connectors of the whole system ensure the stable operation of the system.
5) Real-time accurate display of antenna azimuth, elevation, polarization angle, AGC voltage.
6) Preset and modify complete specifications of 10 satellites, convenient for one-key guiding.
7) Display and modify various parameter settings of antenna control unit and beacon tracking receiving unit.
8) Provides comprehensive tips and alarm information.
9) Provide multiple interlock control to ensure stable and safe operation of the system.
10) Can realize antenna remote control.

2. System Specifications

Item	Description
Appliable antenna	4.5m
Structure type	Integrated

Drive speed	Constant speed
System power consumption	1000W
Angle dispaly range	0～360°
Angle display resolution	0.01°
Encoder accuracy	AZ: ≦0.01°(optional 0.001°)
	AZ: ≦0.01°(optional 0.001°)
	Pol: ≦0.08°(optional 0.01°)
Tracking accuracy	1/10 reception beamwidth
Preset satellite	10 satellites
Remote interface	RJ-45 networks
Working environment	Working temperature: 0℃~40℃. Cable working temperature: -10℃~60℃ (optional: low-temperature cable: -30℃~90℃)
Power supply	Antenna drive: AC 380V±10%, 50/60Hz
Dimension	19inch/3U
Safety	1. Soft and hard limits provide guarantee for the safe operation of the antenna. 2. The cables are all flame-retardant series cables; 3. One-click stop further improves the safety of system use.

Table 2.1 – System Specifications

3. System Diagram and Module Composition

3.1 System Diagram

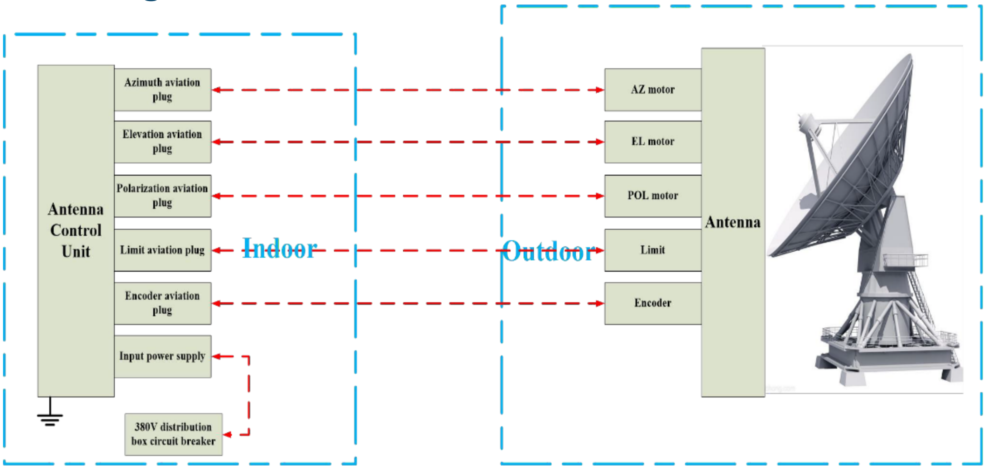

Figure 3.1 – System Diagram

The system has two control modes for the antenna: automatic control mode and manual control mode. The usage and differences of the two control modes are shown in Table 3.1.

Control mode	Control details	Using condition
Manual mode	1. Can directly use the operation to control the antenna movement. 2. The motor rotation hard limit is effective. 3. Press the "Mode" button on the driver panel to enter the manual mode. The interface displays the control mode as "Manual". At this time, user can only operate through the panel buttons, and the touch interface becomes read-only and cannot be operated.	Installation and maintenance Emergency control
Automatic mode	1. Use the touch screen interface to interact with the system control program to complete all antenna control operations. 2. Both the hard limit and soft limit of the motor rotation are valid. 3. Pop up the "Manual" button on the driver panel to enter the automatic mode. The interface displays the control mode as "Local Control".	Daily use

Table 3.1 - Control Mode Description

3.2 System Composition

The antenna control system uses a high-performance processor, which provides a strong guarantee for the realization of various algorithms of the system. The control part, drive part, receiver part and sensor part are integrated, and the operation is simple and practical (Figure 3.2).

3.2.1 Antenna Controller

Figure 3.2 - Antenna Controller

The main hardware configuration of ADU is as follows:

- Power circuit breaker
- Motor forward and reverse module
- Emergency stop switch, one-button stop
- DC power module
- Main control board
- Beacon receiver

Satellite beacon receiver is mainly used to receive single-tone beacons in the L band, accurately estimate their power, and nominalize them in digital or analog voltage.

The product's superior capture time and amplitude stability enable it to simultaneously meet the satellite search applications of portable stations, static communication, and dynamic communication antenna systems. The product appearance is shown in Figure 3.3.

Figure 3.3 - Satellite Beacon Receive

The product has the characteristics of small size, light weight, high reliability, long life, impact vibration resistance, strong anti-interference ability, etc.

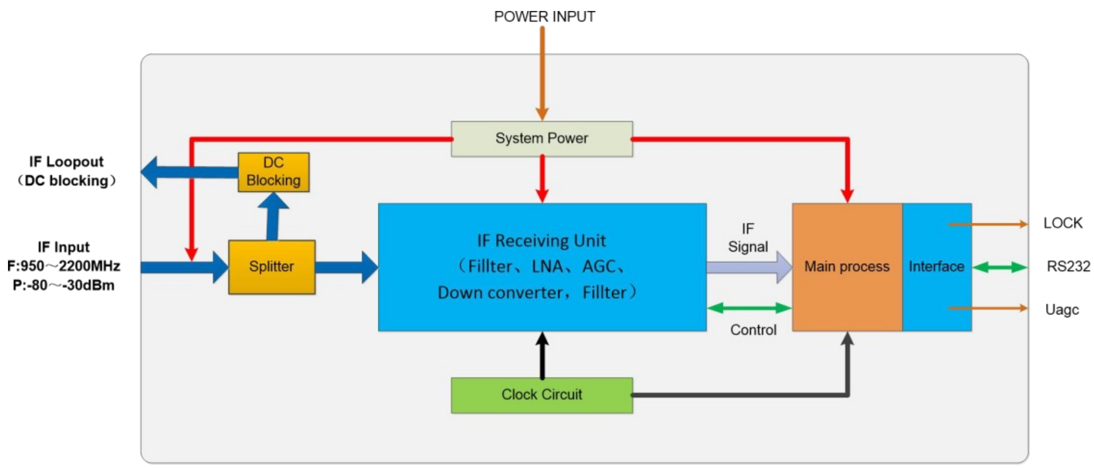

Figure 3.4 - Product Principle

3.2.2 Encoder

The encoder collects the real-time angle of the antenna movement through a code that corresponds to the position and then determines the positive and negative directions and the position of the displacement.

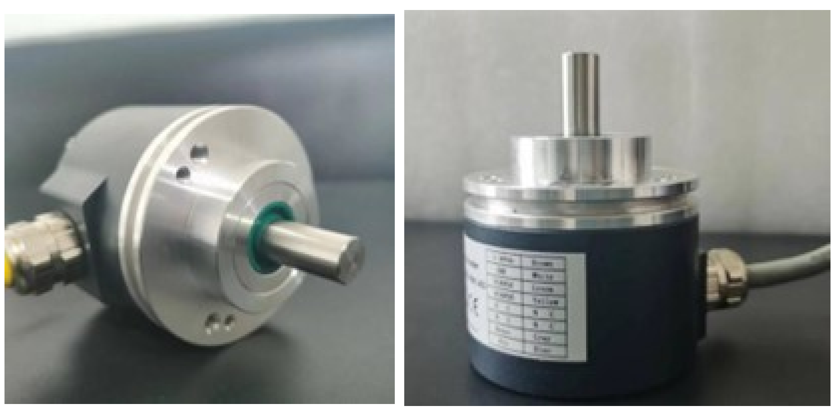

Figure 3.5 - Encoder

3.2.2.1 Limit Switch

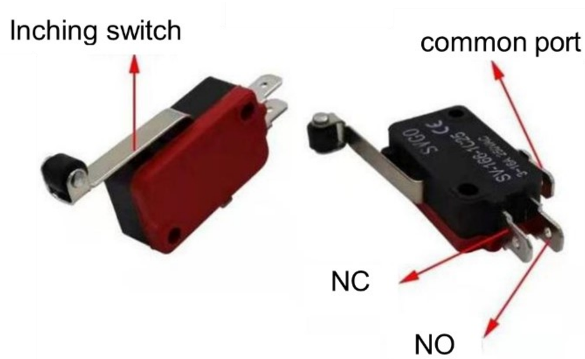

Figure 3.6

4. System Installation and Wiring

4.1 System Installation

Before onsite installation, check whether the devices on the front and rear panels are damaged and whether fixing screws are loose after long-distance transportation. The screws, connectors, and integrated circuits in the chassis are loose or not.

The device has a 3U standard chassis and can be directly installed on the standard rack on site. As shown in Figure 4-1, consider the following factors when selecting an installation position:

- The cable length from the equipment to the controlled antenna should generally not exceed 80 meters. When the distance exceeds 80meters, choose SC-SC fiber optic interface.
- The device should be installed in a position that is easy to operate if frequently operated.
- Keep a certain distance from electronic devices with strong electromagnetic interference ability.

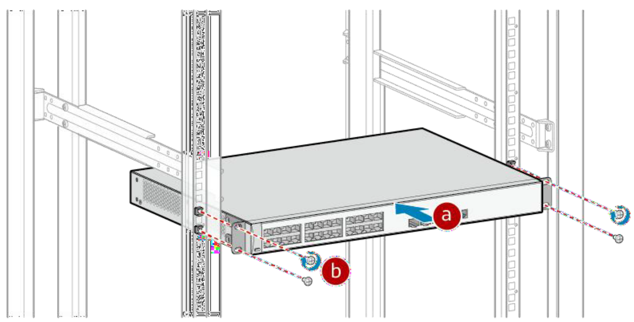

Figure 4.1 - Installation Position

The following factors should be considered:

- When installing the cabinet, keep it as far away as possible from electronic equipment with poor anti-interference ability.
- The equipment must be grounded.

NOTE: All exposed cables must be fitted with protective sleeves, strong current cables and weak current cables must be laid separately during wiring, all cable connectors must ensure reliable, tight, and no soldering connections, and any multi-core or single-core cable must be a complete. The cables are not allowed to be connected in the middle, and there must be no short circuit between any cables.

4.2 Operate and Debugging

Precautions for field debugging

- The installation and debugging personnel must read the manual carefully and have a good knowledge and understanding of the equipment's debugging method, operation and use method and fault handling method.

- Before debugging, it is necessary to confirm whether the drive motor, encoder and mechanical transmission on the antenna are in good condition, and whether the three-phase power supply and grounding meet the use conditions.
- During debugging, operate connectors only when the device is powered off. Otherwise, the circuit or components may be damaged.
- There are many high voltages in the equipment and on the antenna base. Be careful when checking and measuring to avoid electric shock or short circuit that may endanger the safety of the personnel and the equipment.
- During the debugging, the fault detection and protection functions are not normal. Therefore, it is necessary to observe the actual rotation of the antenna when control the antenna to avoid damaging the antenna structure and causing safety accidents and unexpected losses.
- If a fault occurs during debugging, locate the fault cause and rectify the fault according to the fault description in this manual.
- After onsite debugging, record and save all the internal parameters that have been set so that the device can be quickly restored to normal if the stored parameters change or are lost.

4.2.1 Power Supply Check

Before powering the system, check all cables to ensure that they are reliably connected and laid in the specified positions, and cables are not twisted when the antenna rotates.

Observe whether the equipment startup process and panel display are normal. If everything is normal, the debugging work can be started. If it is abnormal, press the emergency stop button to troubleshoot the fault.

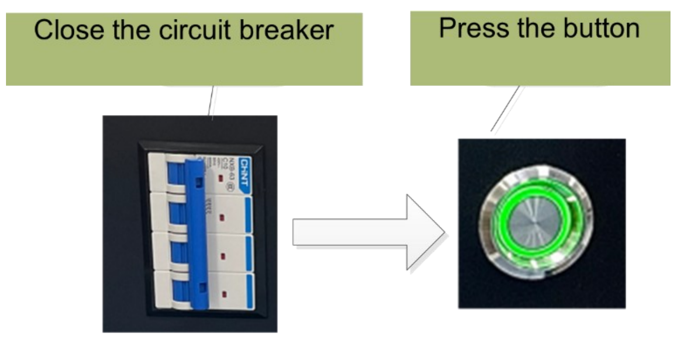

Figure 4.2 – Controller Power Supply

4.2.2 System Debugging

Preparations:

Debugging preparations include understanding the situation and setting relevant parameters. After completing the preparations, plug in other connectors.

Know the situation:

Before field debugging, the following conditions must be known:

- Antenna diameter
- The geographic longitude and latitude of the antenna location
- The bits of the encoder for each axis of the antenna.
- Polarization control, whether the need for polarization control, position detection, motor type, etc.

Process:

A. Adjust antenna and polarization rotation direction

Control antenna movement through manual(control) mode and observe whether the direction of antenna rotation is consistent with the displaying direction of controller. If not, after shutting down the power, swap the wiring of motor U, V or L1, L2, and then restart the power and try again. It should be normal.

Figure 4.3 - ACU Control Panel

Figure 4.4 - Display Interface

B. Adjust the direction of position display change

Control the azimuth movement manually and observe whether the azimuth rotation direction is consistent with the corresponding change direction of the azimuth angle on the screen display interface (when rotating clockwise, the displayed azimuth angle value increases). If not, turn off the power, reverse the encoder, and then turn it on again. Use the same method to adjust the change direction of the pitch and polarization angles.

C. Encoder calibration

After the angle encoder is installed, the angle value displayed is random and cannot represent the real angle of antenna azimuth, elevation and polarization, so the angle must be calibrated. Before calibrating the encoder, the user should control antenna to position at target satellite, when the AGC voltage value of the satellite reaches the maximum, record the satellite's theoretical azimuth, pitch, and polarization angles and fill in the angle calibration position.

D. Commissioning method for hard limit

Turn the CW and CCW limit switches in turn by hand, observe the limit indicator light on the panel, and check whether the limit detection of each switch is normal. If not, check the corresponding wiring and switch to make it normal. Check the upward, downward switch limit and polarization switch limit detection in the same way.

According to the rotation range of the antenna, control the antenna azimuth to the clockwise and counterclockwise limit position respectively, adjust the limit bump block, and rotate the antenna to repeatedly hit the limit several times to confirm that the switch limit is effective. Set the elevation and polarization limit in the same way.

E. Commissioning method for soft limit

The clockwise and counterclockwise soft limit value should be set ahead of the corresponding hard switch limit, and the difference is not less than 2 degrees. In this way, the azimuth has soft limit protection before the switch limit. Adjust the elevation and polarization soft limit in the same way.

After field debugging, all detection and control functions of the equipment should be normal; otherwise, relevant parts should be adjusted to make relevant functions work normally. Then, save all parameters properly so that the device can be quickly restored to normal operation if the stored parameters are changed or lost.

4.3 Operation Description

The control mode (Figure 4.5）divides automatic and manual type.

Figure 4.5 - Control Mode

Manual control can be used temporarily during installation and debugging, and automatic control can be used after system debugging. To ensure stable operation of the antenna, do not touch the manual button at will.

4.3.1 Manual Mode Operation

Press the buttons on the panel to control the movement of antenna. Refer to Figure 4.6.

Figure 4.6 - Controller Panel

Button	Button operation	LED indicator definition
Mode	Press and system enter manual mode, bounce up and system returns to automatic operation mode.	The green light is on in manual mode and off in automatic mode.
AZ CW	Press "CW" and antenna move clockwise, the AZ angle increases, the rotation stops once the button bounce up. Antenna stop rotating when the hardware limit is hit.	The Green is light under the current operation and red will be light when rotation hit limit.
AZ CCW	Press "CCW" and antenna moves counterclockwise, the AZ angle decreases, the rotation stops once the button bounce up. Antenna stop rotating when the hardware limit is hit.	
EL UP	Press "UP" and antenna moves upward, the EL angle increases, the rotation stops once the button bounce up. Antenna stop rotating when the hardware limit is hit.	
EL DOWN	Press "DOWN" and antenna moves downward, the EL angle decreases, the rotation stops once the button bounce up. Antenna stop rotating when the hardware limit is hit.	
POL CW	Press "CW" and polarization move clockwise, the pol angle increases, the rotation stops once the button bounce up. polarization stop rotating when the hardware limit is hit.	
POL CCW	Press "CCW" and polarization moves counterclockwise, the pol angle increases, the rotation stops once the button bounce up. Polarization stops rotating when the hardware limit is hit.	

Table 4.1 Manual mode

NOTE: For example, if you click the elevation “UP”, the antenna will move upward. If you click the elevation “DOWN”, the antenna will not stop moving downward. In this case, user must firstly click the pause button and then click the elevation “DOWN”. The operation methods for other axes are similar.

4.3.2 Automatic Mode Operation

Figure 4.7 - Automatic Mode Interface

The main interface mainly consists of three parts: the top is the system information display bar, the middle is the function main menu bar, and the bottom is the operation information prompt bar.

Menu Options	Description
Sat Guide	Guide antenna to target position as per satellite selection
Position Guide	Guide antenna to the specified position as per the input angle value.
Manual	Realize the different rotation control through button operation in different direction.
Scan/Track	Scan and Track:"Tracke" is based on the strength of the satellite beacon read by the system, the system automatically controls the rotation of the antenna in all directions, so that the antenna locate in the position that has max signal; “Scan” is based on the AZ/EL range set by user, according to the set step and mode within the range to find the maximum signal point within the range.
System Setting	Encoder calibration, local latitude and longitude setting, tracking parameters setting, software limit, satellite information, system time, password management and other functions.
Information	Display the company philosophy, name, contact, website and controller model.

Table 4.2 - Function Menu

4.3.2.1 Sat Guide

①Click “Sat Guide” in the main interface.
②Click "Satellite Selection" button in the Sat Guide box, see Figure 4-8;
③Select the preset target satellite

At this time, the satellite information, beacon frequency and target angle are calculated and displayed.

④Click "Sat Guide" button, and the system will automatically guide the antenna.

Optional: Under this function, the system defaults to high speed when the azimuth and elevation motors are running.

Figure 4.8 - Sat Guide

Figure 4.9 - Satellite Selection

NOTE: Before selecting a satellite in the interface of Figure 4.9, user must first preset the relevant parameters of the target satellite in the main interface settings menu. This system supports parameter presets for 10 satellites.

4.3.2.2 Scan/Track

Scan Operation

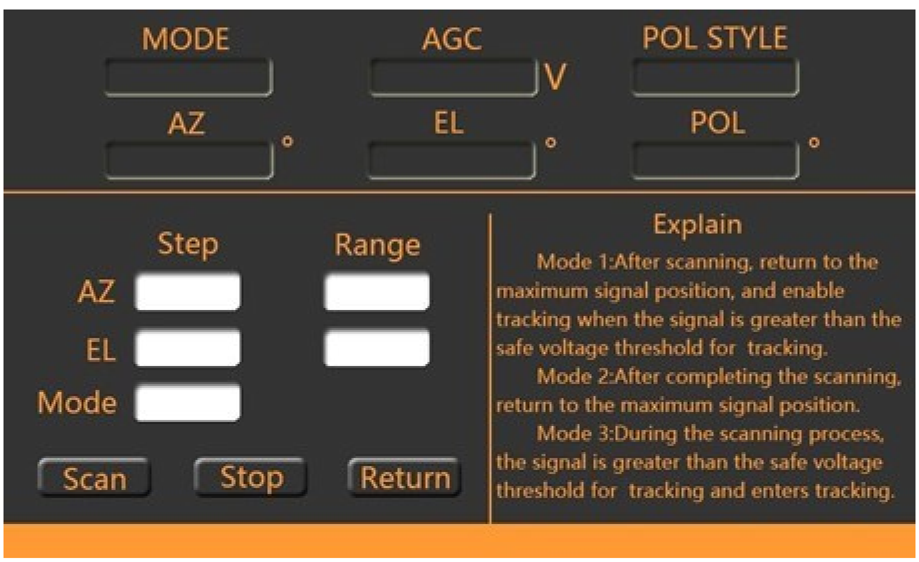

Figure 4.10 - Scan Operation

1) Click “Scan/Track”;
2) Click “Scan/Track” to enter Figure 4-10, input relevant parameters, and select the mode to save;
3) Click Start “Scan/Track” to enter the scanning state. Do not operate during the scanning process.

Automatic Tracking

Figure 4.11 - Automatic Tracking Setting

1) Click “Scan/Track” on the main page;
2) Click Auto Tracking, enter the setting page to set the parameters, and then click Save Parameters as shown in Figure 4.11;
3) Click Start Tracking, the antenna starts to monitor the AGC voltage of the system, and automatically starts and stops the movement of the antenna according to the currently saved maximum value and the tracking parameters in the system settings, so that the antenna is always in the best signal state;

4.3.2.3 Position Guide

The Position Guide operation is shown in Figure 4.12. Enter the target azimuth, elevation, and polarization angles respectively, click to execute the Guide, and the antenna will automatically stop after it reaches the set target angle. (**Optional**: During the operation, the system defaults to high speed when the azimuth and elevation motors are running.)

Figure 4.12 - Position Guide

4.3.2.4 System Setting

Click on the main interface system settings to enter the interface shown in Figure 4.13. Before entering each setting function, the user needs to enter the management password first. The factory default password is **000000**. After the password is verified correctly, each function can be set then.

Figure 4.13 – System Setting

4.3.2.4.1 Encoder Calibration

Figure 4.14 - Encoder Calibration

After the angle encoder is installed, the displayed angle value is a random value, so the angle must be calibrated, as shown in Figure 4.14.

First, point the antenna to a satellite with a known orbital longitude, calculate the actual azimuth elevation of the antenna and the required polarization angle, and complete the calibration after input.

4.3.2.4.2 Local Longitude and Latitude Setting

Simply fill in the local longitude and latitude and click Save, as shown in Figure 4.15.

Figure 4.15 - Longitude and Latitude Setting

4.3.2.4.3 Soft Limit Setting

In order to protect the antenna from rotating out of the normal working range due to unexpected reasons during the rotation process, a software limit position is specially set, and the interface is shown in Figure 4.16. The software limit position is slightly smaller than the hardware limit position of the system.

When the system is running automatically, if the antenna rotates to the software limit, the system will control the antenna to stop immediately for protection. At this time, the antenna should be manually rotated in the opposite direction to release the limit.

Figure 4.16 - Soft Limit Setting

4.3.2.4.4 Tracking Parameters Setting

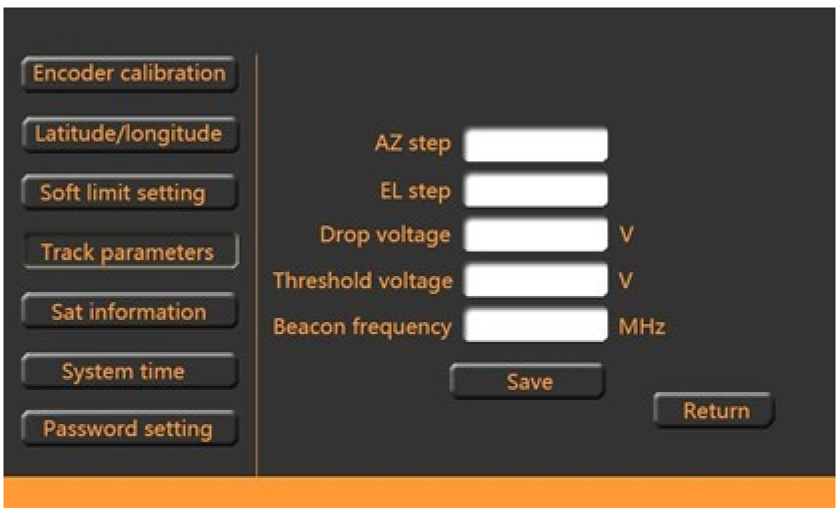

Figure 4.17 - Tracking Parameter Setting

Click the Tracking Parameters button to enter the interface shown in Figure 4.17, and set: azimuth step, elevation step, drop voltage, threshold voltage, beacon frequency. If the input is correct, click OK to complete the setting.

4.3.2.4.5 Sat Information

In a new system, some satellite information is usually added in advance for direct access. If the expected satellite is not in the added part, user can add or delete it according to the information.

How to add satellites:

- Click Satellite Information on the main interface to enter the preset satellite information page as shown in Figure 4.18;
- Click “Sat information” to add new satellites according to the information on the page;
- After input, press the "OK" button to save the result.

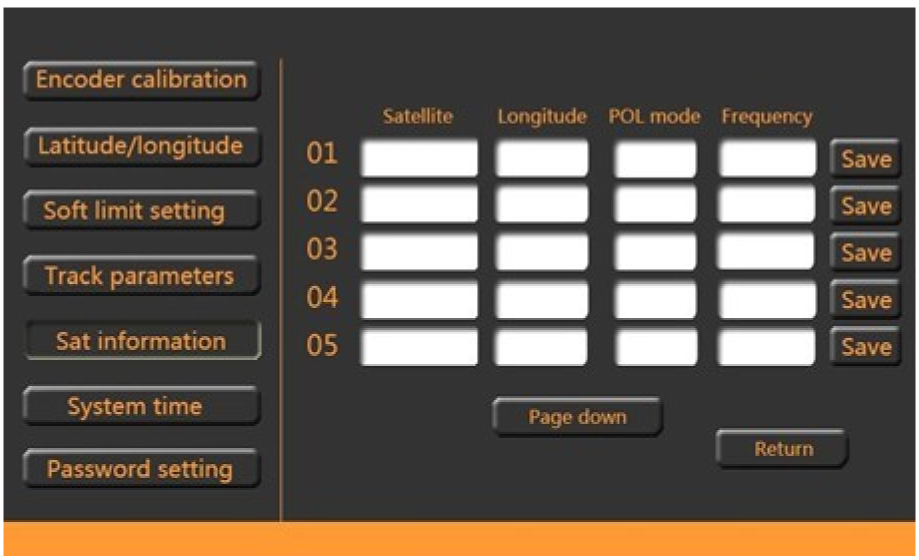

Figure 4.18 - Preset Satellite

4.3.2.4.6 System Time

Click System Time to enter the interface shown in Figure 4.19. Time information of this system can be modified according to the prompts on the interface.

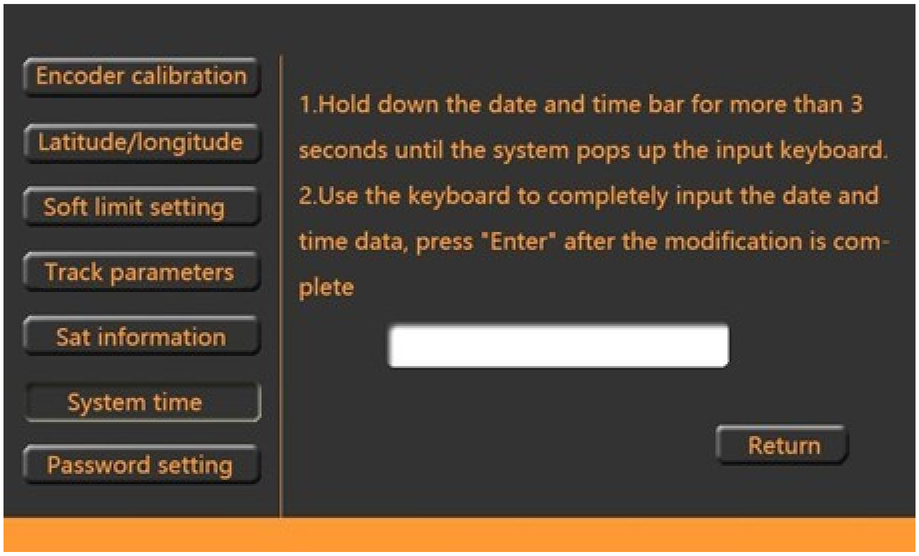

Figure 4.19 - System Time

4.3.2.4.7 Password Resetting

Click Password setting to enter the interface as shown in Figure 4.20. According to the prompts on the interface, user can modify the password of this system.

Figure 4.20 - Password Resetting

4.3.2.4.8 System Information

Click the “Information” on the main interface to enter the interface shown in Figure 4.21.

Figure 4.21 - System Information

5. Troubleshooting and Maintenance

5.1 Troubleshooting

When the equipment is in normal operation, it can automatically diagnose antenna soft and hard limit and encoder angle faults, and display and alarm the detected faults, to facilitate equipment maintenance.

Types	Troubleshooting methods
Soft limit	When the antenna is in soft limit, use manual mode to control the antenna to exit the corresponding limit.
Hard limit	1. Check whether there is a corresponding soft limit alarm before the hard limit. If not, it indicates that the soft limit is not properly set and should be reset. 2. When the antenna is in hard limit, use manual mode to control antenna to exit the corresponding limit. 3. If the antenna is not in the limit position when the hard limit alarm occurs, check whether the limit switch is damaged and whether the connection cable is normal.

Angle display failure	1. Check whether the corresponding encoder and its connecting cable are normal. 2. Adjust the direction of encoder. 3. Check the bits of the encoder.
Motor rotation failure	1. Check whether the corresponding motor and its connection are normal. 2. Exchange any two three-phase lines of U, V and W of azimuth and elevation motors. 3. Exchange L1, L2 cable of polarization motor.
Abnormal power/voltage	1. Check whether the three-phase power supply is missing phase. 2. Check the related AC power failure and check the circuit.

Table 5.1 - Troubleshooting Methods

5.2 Maintenance Interface Settings

Click the main interface system settings, enter the super password in the interface, the factory password is 5288. After the password verification, the function setting can be performed, see Figure 5-1, there are 5 setting options on the left side, namely: Feed, MULTIAXIS(motorized polarization), encoder bits, encoder reverse, and Beacon.

Figure 5.1 - Setting

5.2.1 LNB Voltage Setting

Click FEED button to display the interface as shown in Figure 5.2. In this interface, the user can select an appropriate voltage to supply power to the LNB.

Figure 5.2 - LNB voltage Setting

5.2.2 Motorized Polarization Setting

Click MULTIAXIS button to display the interface as shown in Figure 5.3, where the number of antenna axes can be selected.

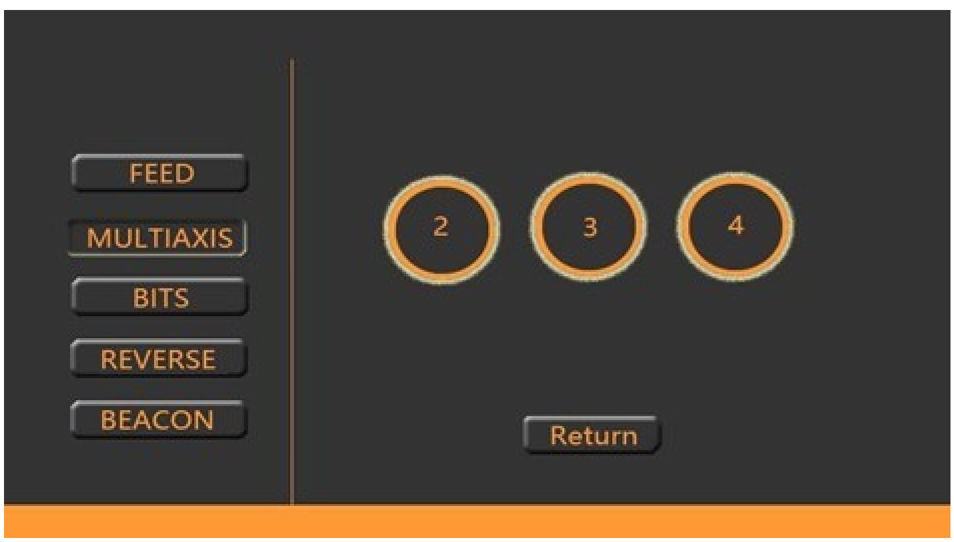

Figure 5.3 - Motorized Polarization Setting

5.2.3 Encoder Bit Setting

After the antenna installation is completed, the system needs to perform initial settings, click BITS, enter the interface as shown in Figure 5.4, input the current antenna azimuth, elevation, and polarization encoder digits in turn, and click OK.

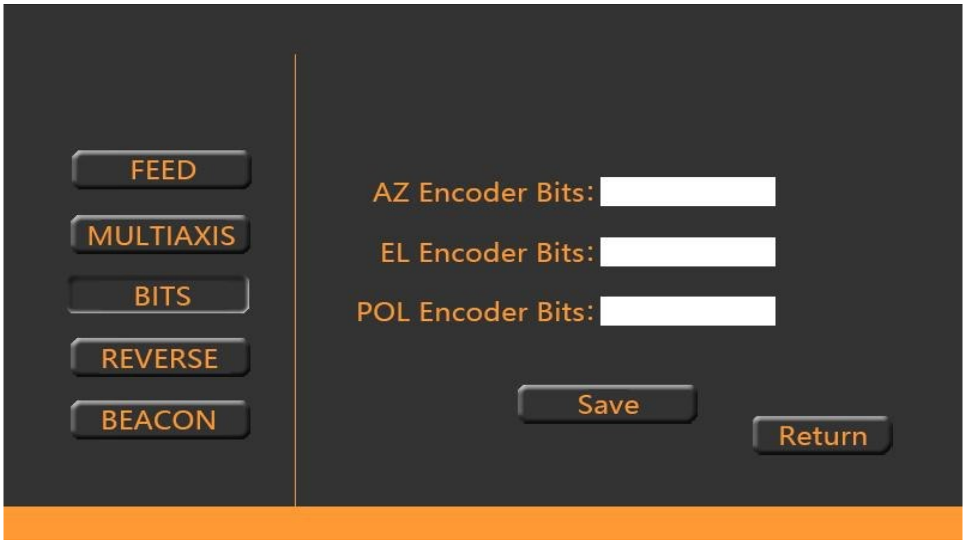

Figure 5.4 - Encoder Bit Setting

5.2.4 Encoder Reverse Setting

When the antenna is being debugged, it may encounter the elevation upward movement, but the elevation angle display decreases, which means that the encoder direction is opposite to the antenna rotation direction. At this time, the encoder needs to be set direction reverse, that is, when elevation rotate upward, and its angle display is increased. Click REVERSE to enter the interface in Figure 5.5 and select the correct direction of the encoder to set it.

Figure 5.5 - Encoder Reverse Setting

5.2.5 Receiver Setting

Click BEACON to enter the interface of Figure 5.6, and the user can set the parameters of 22K, gain and slope.

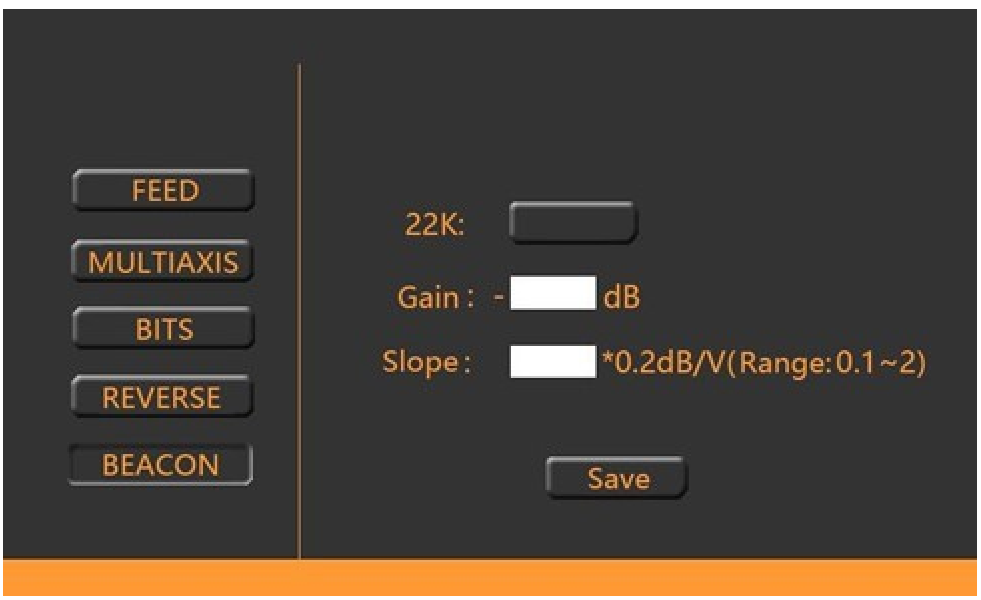

Figure 5.6 - Receiver Setting

6. System Maintenance

The purpose of system maintenance is to prevent abnormal device operation, performance deterioration, and device failure, and ensure personal safety and normal operation of electrical devices. The following six methods are generally used for device maintenance. Look: observe the abnormal appearance of components of electrical equipment. Such as: fasteners are loose, insulation materials are carbonized black; Observe the value indicated by the instrument or the state presented by the indicating device. Listen: listen the sound changes of electrical equipment to judge the working condition. For example, asynchronous motor can not started and issue "buzzing" sound; When motor bearing damage and issue "rustling" sound. Smell: Smell the odor of electrical equipment when it is in operation. Such as electrical equipment due to short circuit, overload and other faults caused by temperature rise beyond the limit, can appear pungent burnt taste. Touch: by touching the temperature of the shell of electrical equipment to roughly judge whether the operation of low-level insulating equipment or general equipment is normal. Test: test various operating parameters and insulation resistance values of electrical equipment with common measuring instruments. Do: according to the electrical equipment maintenance requirements of regular cleaning and maintenance, inspect and maintenance.

- Periodically check whether the operating switches and buttons of the device are normal. If there is burning phenomenon, handle it in time.
- Check all circuit connections regularly for looseness, especially oxidation. Pay special attention to those phenomenon and deal with in time.
- Regularly remove dust and oil on equipment, circuit and motor to keep equipment clean.
- Periodically check whether the grounding cable is in good condition and measure the grounding resistance of the grounding cable. The grounding resistance must meet the requirements (no more than 4 OHms). Otherwise, find out the cause and try to reach the specified value.

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